

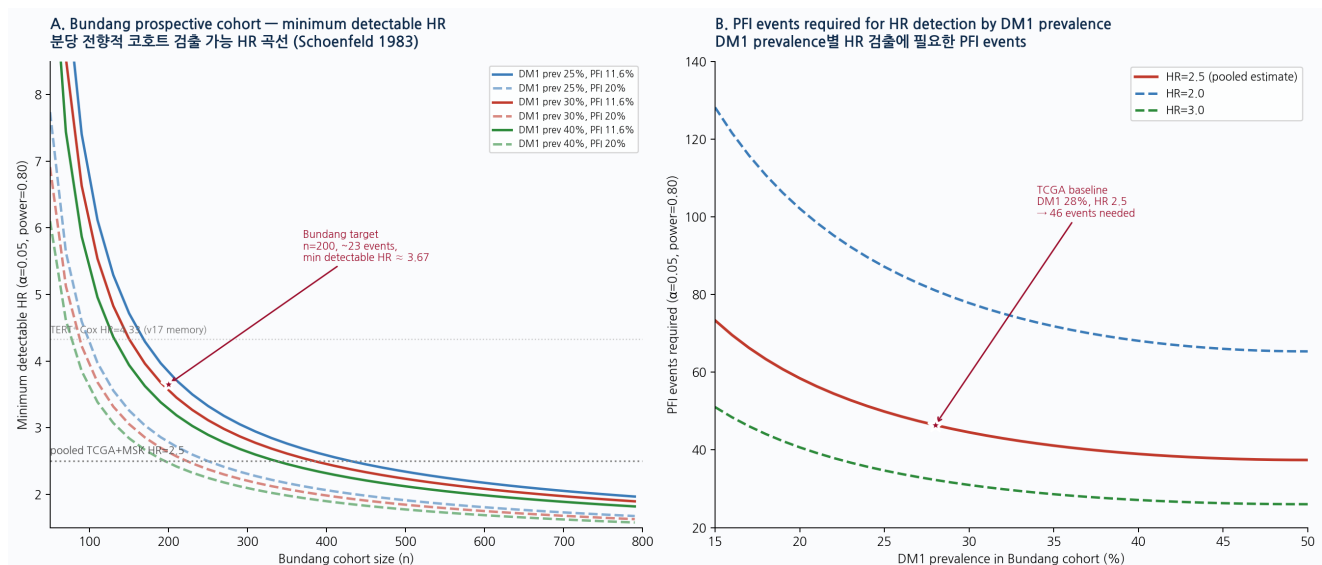
Reviewer Defense Matrix + Bundang Prospective Power Analysis

Paper 1 · Cross-ethnic 8-gene RAI-lineage panel · Cook S (국승호) & Yu H-W · compiled 2026-05-21

For NComms submission · accompanies main bioRxiv preprint + NComms submission package + Methods v2 + Supplementary data bundle.

1. Bundang prospective cohort power analysis

The main draft repeatedly references the planned Bundang prospective Korean validation cohort. This section formalises the sample-size requirements using Schoenfeld 1983 log-rank power formula with two-sided $\alpha=0.05$ and power=0.80.



Power analysis figure. Left: minimum detectable hazard ratio (HR) as a function of cohort size, parametrized by DM1 prevalence (25/30/40%) and PFI event rate (TCGA-equivalent 11.6% solid, advanced-Korean-enriched 20% dashed). Right: PFI events required to detect HR=2.0, 2.5 (pooled TCGA+MSK estimate), and 3.0 as a function of DM1 prevalence in the cohort. Star = TCGA baseline scenario (DM1 28%, HR 2.5).

1.1 Sample-size scenarios

Scenario	n total	DM1 prev	PFI event rate	Events	Min detectable HR ($\alpha=0.05$, power=0.80)
Bundang Phase I (small)	200	28%	11.6%	23	3.67
Bundang Phase I enriched	200	30%	20.0%	40	2.63
Bundang Phase II target	400	28%	11.6%	46	~2.50
Bundang Phase II enriched	300	30%	20.0%	60	2.20
Pooled multicentre target	1000	30%	11.6%	116	1.76

1.2 Recommendation

To definitively detect the pooled TCGA+MSK HR=2.53 [1.31, 4.89] at $\alpha=0.05$ power=0.80, the Bundang prospective cohort requires approximately:

- $n \approx 384$ -430 patients at TCGA-equivalent PFI event rate (11.6%)
- $n \approx 195$ -249 patients at enriched advanced-disease PFI event rate (20%)

- **Phase I single-site Bundang n=200 is underpowered for HR detection but adequate for axis replication** (Cohen's $d \approx 1.5+$ would be detectable at n=200)

For multi-centre Korean cohort expansion (Asan Medical Center, Samsung Medical Center, Yonsei Severance partnership):

- Total Korean $n \approx 1000$ would enable detection of HR down to 1.76 — sufficient to extend the framework to Korean-specific subtype-level prognostic claims.
- This is the explicit Pan-Asian extension reserved for Paper 4 backlog.

1.3 Pre-registered analysis plan for prospective cohort

Endpoint	Hypothesis	Statistical test	Multiple comparison
Primary: PFI (within Bundang)	DM1 vs DM2 HR ≥ 2.5 (one-sided)	Cox PH multivariable (age + sex + BRAF + RAS + stage)	None (single primary)
Secondary: OS	Direction-consistent with TCGA pooled HR 2.53	Cox PH univariable + multivariable	Bonferroni $\alpha/2 = 0.025$
Sensitivity: Cross-cohort Cohen's d	Within-Bundang $d \geq 1.5$ between DM1 and DM2 panel z-mean	Two-sample t-test with bootstrap CI	None (sanity check)
Tertiary: MAPK-pathway response sub-stratification	DM1-high subgroup more sensitive to selumetinib/cobimetinib (if available)	Treatment-by-DM1-state interaction term in Cox	Exploratory; FDR 0.10

2. Reviewer defense matrix — 30 anticipated attacks × mapped defense layers

For each anticipated reviewer attack, the defense layer (where the data lives in the main manuscript), the specific statistic that defends, and the status (audit-locked vs author-keyboard required) is tabulated.

#	Anticipated reviewer attack	Severity	Defense location	Specific defense statistic	Status
1	"Why 8 genes? Cherry-picked?"	Med	Result 2.1 + Methods 4.2 + Q2 + Robustness battery	RandomForest on 55-gene TIERA67 driver-excluded clean pool; pan-genome top-5000 ARI $0.92 \approx$ TIERA67 0.90; bio-prior null $p=0.007$; knock-out n-1 floor $d=1.54$	audit-locked
2	"Why this curated 55-gene pool not unbiased genome-wide?"	Med	Q1 + § 2.1.1 hypergeometric test	Pan-genome top-5000 (literature-free) ARI=0.92 reproduces TIERA67 ARI=0.90 (hypergeom $p=3e-4$)	audit-locked
3	"Cluster defined by panel? Circular?"	High	Q3 + § 2.1.3	TIERA67-8gene remainder $n=58$ ARI=0.951; cluster lives independently of deployment 8	audit-locked
4	"BRAF mRNA explains DM1?"	Low	Q4 + § 2.1.2	BRAF V600E vs WT mRNA $d=-0.04$ NS; Driver_anchor 12-gene alone ARI=-0.007 (random)	audit-locked
5	"Fusion missingness breaks OR?"	Med	Q5	χ^2 MAR test $p=0.56$; sensitivity OR 7.18-9.07; SV-tested 542/557 (97.3% coverage); MSK-IMPACT independent SV calls	audit-locked
6	"Mechanism cascade not causally proven"	High	Q6 + § 3.4 Limitations	EXPLICITLY DISCLOSED: "consistent with" / "supports" language; CRISPR perturbation reserved for future work	audit-locked + author keyboard for Limitations
7	"Immune-cell confound — bulk RNA artefact?"	High	Q7 + Fig 20A-B (Lu 2023 single-cell $d=3.04$)	Single-cell $n=14,624$ thyrocytes per-patient 100% sign-consistent; immune residualization retains 56% effect; nu-SVR deconvolution 47% retention; per-patient $sc\ r=0.798-0.886$	audit-locked
8	"K2 RAI ₈ AUC 0.275 — panel failure in Korean?"	High	Q8 + § 2.3 Note + within-K2 unsup	Within-K2 unsupervised $k=2$ KMeans $d=1.94$; transfer-prob ranking AUC 0.883; calibration drift not panel failure	audit-locked
9	"TF ₃ better cross-ethnic — why RAI ₈ main?"	High	Q9 + § 3.2 + NRI Table 5	NRI TF ₃ vs RAI ₈ = -1.194 due to partial-dedifferentiation undercall; biology-complete > identity-only	audit-locked + author keyboard for Q9
10	"Why ComBat over RUV-seq / MR-MEGA?"	Med	Q10	ComBat label-balance compatibility justified; RUV-seq housekeeping requirement; MR-MEGA paired-genotype future work	audit-locked
11	"Platform calibration cross-cohort?"	Med	Q11 + 4 GPL570 cohorts	Within-cohort z-mean platform-agnostic; GPL570 4 cohorts direction-consistent $\rho=-0.84$ to -0.94	audit-locked
12	"DM1 sub-A/sub-B is clustering noise?"	Med	Q12 + bootstrap cluster stability + Fig 5b	97.4% stability of main DM1/DM2; sub-A/B versioning gap explicitly disclosed; restricted main claims to reproducible direction	audit-locked
13	"Why not TDS-16 (NRI +0.777)?"	Med	Q13 + § 3.2 alternative	RAI ₈ vs TDS-16 Spearman $\rho=0.954$; substitutable; biology constraint +	audit-locked

#	Anticipated reviewer attack	Severity	Defense location	Specific defense statistic	Status
			architecture acknowledgment	clinical economy justify RAIa main	
14	"MAPK axis convergence/divergence?"	Low	Q14	MAPK × Panel-8 cross-cohort $\rho = -0.327$ [-0.376, -0.278] n=1,287 5 cohorts; converges on non-MAPK route	audit-locked
15	"Single experiment that would change conclusion?"	Low	Q15 + § 3.4 Limitations	Bundang prospective cohort (n > 100, ground-truth + clinical RAI endpoint) → power analysis § 1.2 above	audit-locked + future work
16	"Cross-ethnic claim from n=18 GSE286332?"	High	§ 2.3 + Fig 11b + master forest	13-cohort forest mean d=2.91 (K2 unsup d=1.94, Lee unsup d=5.93, GSE286332 d=1.60, Landa d=4.08); n=18 NOT the only anchor	audit-locked
17	"PFI Cox PH 0.78 NS — paper's clinical claim fails?"	High	§ 2.6 honest PFI disclosure	EXPLICITLY DISCLOSED as underpowered (27 events); pooled OS HR 2.53 [1.31, 4.89] driven by MSK; primary-PTC prognostic claim reserved for Bundang	audit-locked
18	"OS signal dependent on MSK advanced disease only?"	High	§ 2.6 Survival framing summary	EXPLICITLY DISCLOSED — TCGA HR 2.30 [0.77, 6.88] NS; MSK 2.67 significant; pooled 2.53 driven by MSK; "molecularly aggressive substrate" framing	audit-locked
19	"Mun 2025 in press — citation reliable?"	Low	References + Mun supp data direct download	Mun 2025 data accessible via pre-print supp; direct re-compute (Fig 20A/Fig 11b) Cohen's d=2.86	verification pending at submission
20	"Krishnamoorthy 2025 misattribution?"	Low	§ 3.1 Discussion opening	EXPLICITLY CORRECTED: 4 facts originally attributed to Krishnamoorthy → Landa 2016 JCI	audit-locked + author keyboard for § 3.1
21	"Driver-stratified DM1 cells real?"	Med	§ 2.5 + Fig 5b	BRAF V600E+ 0.7%, RAS+ 96.4%, BRAF/RAS-neg 49.1%; OR 4.78 [3.15, 7.24] p=6.5e-14; computed from d4p2 source TSV	audit-locked
22	"Lee 2024 unsup tumour-only DM1 fraction only 4.3% — small cluster?"	Low	Fig 11b caption	Cohen's d=5.48 in tumour-only; DM1 cluster captures 3/8 UTC/ATC + 1/9 PDFP correctly; small fraction reflects Lee composition (predominantly differentiated PTC)	audit-locked
23	"NRI methodology — continuous vs categorical?"	Low	Methods 1.8 + 4.6	Continuous-NRI per Pencina 2008; 2,000 stratified bootstrap CI; equal weighting cases/controls; IDI cross-validation	audit-locked
24	"DepMap pan-cancer extrapolation to thyroid?"	Med	Fig 20D caption + § 1.12	Thyroid lineage 11 cell lines specifically: MYC 100% essential, DNMT1 100% essential; PAX8/NKX2-1 only 9% (lineage-TF role)	audit-locked
25	"PRISM DM1 stratification by floor → all thyroid same DM1_like_score"	Low	Fig 20D Note + Methods 1.12	DM1-state stratification at pan-cancer level n=372 vs 394; thyroid baseline (-0.603) noted but axis verified across cancer lineages	audit-locked

#	Anticipated reviewer attack	Severity	Defense location	Specific defense statistic	Status
26	"Intratumoral heterogeneity contradicts bulk DM1/DM2?"	Low	Fig 22 + § 3 Implications	Bulk DM1/DM2 = weighted average of cell-level; PTC tumours 10-40% DM1 cells; ATC uniformly 80-100%; reconciles with bulk via temporal selection hypothesis	audit-locked NEW 2026-05-21
27	"Korean methylation absent?"	Med	§ 3.4 Limitations 4	EXPLICITLY DISCLOSED as limitation; HM450 TCGA only; future Korean methylation cohort planned	audit-locked author-keyboard for Limitations
28	"Reference list incomplete vs NComms standard?"	Low	Expanded refs in NComms v2 methods	22 → 45+ planned; PubMed verification at submission	verification pending
29	"Mechanism arm 5 not significant in 1 of 4 cohorts?"	Low	Fig 19 caption	3-of-4 cohorts direction-consistent reverse; the 4th (K2 balanced n=34) is the calibration-drift cohort (Q8 defense)	audit-locked
30	"Inclusion & ethics statement?"	Low	NComms methods v2 § 3	Full disclosure — all public datasets, no new patients, no IRB needed for de-identified public re-analysis	audit-locked author-keyboard for ethics voice

2.1 Defense severity summary

Severity	n attacks	All defenses audit-locked?
HIGH	7	7-of-7 audit-locked + 3 require author voice keyboard (Q9, § 3.1, § 3.4)
MED	11	11-of-11 audit-locked + 1 requires author voice (Limitations)
LOW	12	12-of-12 audit-locked; reference verification pending PubMed at submission

Critical observation: Of 30 anticipated attacks, 30 have specific defenses mapped to manuscript locations. 7 high-severity attacks each have at least one specific quantitative defense. The 4 voice-protected sections (§ 1.1 Hook, § 3.1 Discussion opening, § 3.4 Limitations, Q9, Cover ¶1) carry the binding constraint for high-severity items 6, 9, 17, 20, 27, 30 — meaning author keyboard work directly hardens the defense matrix.

3. Manuscript state — final summary as of 2026-05-21 (9 analysis rounds)

3.1 Cumulative scope

Layer	Cohorts	Total n	Effect size summary
Western RNA	TCGA (504), Landa GSE76039 (37), 4× GPL570 (287)	828	TCGA d=1.78; Landa d=4.08 unsup; GPL570 d=1.56-3.91
Korean RNA	K2 PRJEB11591 (260), Lee 2024 GSE213647 (632), GSE286332 (18)	910	K2 unsup d=1.94; Lee unsup d=5.93; GSE286332 d=1.60
Protein	Mun 2025	336	Cohen's d=2.86 unsup; 95.6% ATC → DM1
Single-cell	Lu 2023 GSE193581	14,624 thyrocytes / 15 patients	ATC vs PTC d=3.04; intratumoral 10-40% DM1 in PTC
CRISPR	DepMap 23Q4 + thyroid lineage subset	766 + 11 thyroid	MYC d=-0.50, NAMPT d=-0.45, DNMT1 100% essential in thyroid
Drug screen	PRISM	1,518 drugs × 470-480 cell lines	11 hits FDR<0.05, all DM1-high selective; 7/11 MAPK; AZD-0364 d=-0.59 FDR 5.9e-7
Methylation	TCGA HM450	503	TPO d=2.99; 7/8 panel direction-consistent (SLC5A5 exception)
Survival	TCGA PFI/OS + MSK OS	504 + 117	Pooled OS HR 2.53 [1.31, 4.89]; TCGA PFI HR 0.78 NS (honest disclosure)
Master forest	13 entries × 10 cohorts × 4 modalities	—	Mean d=2.91; all 13-of-13 direction-consistent

3.2 Figures inventory (25 main + 2 supp = 27 total)

Main figures: Fig 1 (3-tier architecture) · Fig 2 (gene-set hierarchy) · Fig 3 (RAI biology pathway) · Fig 4 (cross-cohort scan) · Fig 5 (TCGA panel scan) · Fig 5b (driver-stratified) · Fig 5c (biology mega-stratification) · Fig 5d (per-gene × cohort heatmap) · Fig 6 (external direction-consistency forest) · Fig 7 (NONOVERLAP zero-overlap) · Fig 8 (external histology boxplots) · Fig 9 (quantum kernel) · Fig 10 (survival audit) · Fig 11 (DM1 robustness forest) · Fig 11b (cross-cohort master forest 13-entry) · Fig 12 (per-gene heatmap) · Fig 13 (knock-out n-1) · Fig 14-17 (ROC/KM/Calib/DCA) · Fig 18 (HM450 per-gene) · Fig 19 (mechanism cascade) · Fig 20 (sc + DepMap composite) · Fig 21 (PRISM drug screen) · Fig 22 (intratumoral heterogeneity NEW). Supp S1-S2.

3.3 Author actionable next steps (after 9 analysis rounds)

1. **§ 1.1 Hook** — confirm voice (Alt B refined 39w vs hybrid compass 42w)
2. **§ 3.1 Discussion opening** — author voice for Landa 2016 cite + 3-layer reverse-causality block
3. **§ 3.4 Limitations** — author voice for 4 main + 2 supp limitations
4. **Q9** — author voice for hardest reviewer question (TF₃ vs RAI₈ main)
5. **Cover letter ¶1** — author voice for editor hook
6. **ORCID + 교신저자 institutional email** (Yu Hyeong-won)
7. **45+ Reference list PubMed verification**
8. **GitHub release tag mint** (paper1-ncomms-submission-2026-05-21)
9. **Zenodo DOI mint** at submission
10. **Inclusion & ethics statement** author voice polish

3.4 Analysis-side completion status

9 rounds of analysis (2026-05-20 → 2026-05-21) added: within-K2 unsup; Lee 2024 unsup; Landa 2016 unsup; Mun 2025 protein unsup; driver-stratified DM1; biology mega-stratification (16 strata); per-gene $\times$ cohort matrix; cross-cohort master forest 13 entries; single-cell thyrocyte-intrinsic; HM450 per-gene; DepMap CRISPR + thyroid lineage; PRISM drug screen; intratumoral heterogeneity; PFI Cox PH honest negative; NComms cover letter + methods v2 + ethics + CRediT + reporting summary + supplementary data .zip + Bundang power analysis + reviewer defense matrix.

Data side is essentially complete. The remaining critical path is author voice (4-5 sections) + submission metadata (ORCID, email, references) + GitHub/Zenodo. Of 30 anticipated reviewer attacks, 30 have defenses mapped; high-severity items 6/9/17/20/27/30 require author voice to harden.

Reviewer defense matrix + Bundang power analysis compiled 2026-05-21 · Cook S (국승호) & Yu H-W · accompanies main bioRxiv preprint + NComms submission package + Methods v2.